Project Report

On

CDAC STUDENT MANAGEMENT SYSTEM

Submitted in partial fulfillment for the award of

Diploma in Advance Computing (DAC) from

C-DAC, Hyderabad (Ameerpet)

Guided by:

Mrs. Soumya Mam

Presented by:

Sourabh Agarwal | PRN Number 260250320054

Devendra Injapure | PRN Number 260250320016

Shailesh Bhure | PRN Number 260250320056

Keerthana Mavuri | PRN Number 260250320030

Centre for Development of Advanced Computing (C-DAC), Hyderabad

Ameerpet, Hyderabad

ACKNOWLEDGEMENT

This project "CDAC STUDENT MANAGEMENT SYSTEM" was a great learning experience for us and we are submitting this work to the Advanced Computing Training School, C-DAC, Ameerpet, Hyderabad.

We are very glad to mention the name of our guide for the valuable guidance provided to work on this project. The guidance and support helped us overcome various obstacles and intricacies during the course of the project work.

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From:

Sourabh Agarwal (PRN 260250320054)

Devendra Injapure (PRN 260250320016)

Shailesh Bhure (PRN 260250320056)

Keerthana Mavuri (PRN 260250320030)

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1. Introduction of Project

We have developed a CDAC Student Management System — a web-based portal used by C-DAC, Hyderabad (Ameerpet) to manage students, faculty, courses, attendance, marks, assignments, placements, and communication between students and staff for its diploma programs, including DAC, DBDA, DESD and DITISS.

The platform provides two dedicated portals — a Student Portal and a Staff/Faculty Portal. The front end is built using React and Bootstrap for a responsive and user-friendly interface. The back end is powered by Node.js and Spring Boot, exposing secure REST APIs, with MySQL used as the relational database for persistent storage.

Students can log in to view their profile, check assignments, submit assignment work, track attendance, view marks and academic grades, monitor their current CGPA and modules cleared, browse the notice board, explore active placement drives, and submit feedback.

Faculty and staff can log in to manage course modules and curriculum, and to administer core academic services such as student records, marks and grades entry, assignment creation and evaluation, attendance marking, placement drive management, notifications/notice board postings, and reviewing student feedback. The staff dashboard also gives a consolidated view of enrolled students, active placement drives, average attendance, and course-wise statistics across DAC, DBDA, DESD and DITISS.

REST APIs are used extensively across the system, allowing the React front end to communicate securely with the Spring Boot and Node.js services. Authentication is handled using registered institute email IDs for both students and staff, with role-based access separating the student and staff experiences.

2. Product Overview and Summary

2.1 Purpose

Managing student records, attendance, marks, assignments, and placement activity manually across multiple diploma courses (DAC, DBDA, DESD, DITISS) at a busy training centre like C-DAC Hyderabad is time-consuming and error-prone. The CDAC Student Management System automates these processes, giving students self-service access to their own academic data and giving faculty a single dashboard to manage courses, students, and communication.

2.2 Scope

The CDAC Student Management System covers end-to-end academic administration for a training centre: student onboarding, assignment distribution and submission, attendance tracking, marks and grade entry, CGPA computation, placement drive management, notice board communication, and feedback collection — scoped specifically for C-DAC Hyderabad's Ameerpet centre and its DAC, DBDA, DESD and DITISS courses.

2.3 Overview

Section 3.0 (Overall Description) provides an overview of the system's components, technology stack, and the relationship between the student and staff modules. Section 4.0 covers the Functional and User Interface Requirements, including use-case diagrams for both user roles. Section 5.0 describes the High-

Level Design and Database (ER) Design. Section 6.0 presents the user interface screens. Section 7.0 documents the test report, followed by the project management methodology and future scope.

2.4 Feasibility Study

Feasibility is the determination of whether a project is worth doing. Before recommending and building the system, it was evaluated for feasibility in the following ways:

- Technical Feasibility
- Operational Feasibility

Technical Feasibility: The system analyst verified whether it is possible to develop the requested system given availability of manpower, software, and hardware. The system runs on any modern browser across Windows and Linux platforms, using widely adopted, well-documented open-source technologies (React, Bootstrap, Node.js, Spring Boot, MySQL), making it technically feasible without requiring specialised infrastructure. A proof of concept was implemented to verify REST API communication between the React front end and the Spring Boot/Node.js back end.

Operational Feasibility: The proposed system is operationally feasible because both students and staff at C-DAC, being regular users of computers and web applications, can understand and use the system with minimal training. The interface follows familiar dashboard and form-based patterns already common in similar learning management portals.

3. Overall Description

3.1 Product Features

The project's aim is to provide a centralized, web-based Student Management System for C-DAC Hyderabad that is platform-independent, built using React, Bootstrap, Node.js, Spring Boot and REST APIs, covering student academics, attendance, assignments, placements, and staff administration.

3.2 Technology Used

FRONT END

- React — component-based UI library for building the Student and Staff dashboards
- Bootstrap — responsive grid system and pre-styled UI components
- REST client calls (Axios/Fetch) for consuming back-end APIs

BACK END

- Node.js — lightweight REST services for notifications, notice board and feedback modules
- Spring Boot — core academic services: students, marks, grades, assignments, attendance and placements
- MySQL — relational database for storage of all academic and administrative data

React

React is a JavaScript library for building user interfaces. It uses a Virtual DOM to efficiently update only the components that change, which improves performance for data-heavy dashboards such as the student and staff home screens used in this system.

Bootstrap

Bootstrap provides a responsive, mobile-first grid system and a library of pre-built UI components (navigation bars, cards, tables, forms, buttons), which speeds up development of a consistent, professional-looking interface across the login, dashboard, and management screens.

Spring Boot

Spring Boot is used for rapid development of Java-based REST APIs with an embedded Tomcat server, eliminating the need for external application servers. It powers the core academic modules such as marks, grades, attendance, assignments and placements.

Node.js

Node.js is used to build lightweight, event-driven REST services for the notice board, notifications, and feedback modules, complementing the Spring Boot services for the rest of the system.

MySQL

MySQL is an open-source relational database management system (RDBMS) based on SQL. It is used here to store all student, staff, course, attendance, marks, assignment, placement, notice and feedback data, enabling fast, structured queries across modules.

3.3 User Classes

There are two types of users who can access this system:

- Student — can view their profile, assignments, attendance, marks, grades, CGPA, modules cleared, placement drives, notice board, and can submit feedback.
- Staff / Faculty — can manage students, marks, grades, assignments, attendance, placement drives, course modules & curriculum, post notifications, and view student feedback.

3.4 General Constraints

- Users (students and staff) must have a valid C-DAC institute email ID and access to a web browser.
- Stable internet connectivity is required to access the portal.
- Student and staff data is scoped to C-DAC Hyderabad (Ameerpet) and its DAC, DBDA, DESD and DITISS courses.

4. Requirements

4.1 Functional Requirements

4.1.1 Complete System — Student Module

Student use cases: Login/Sign In, My Profile, View Assignments, Submit Assignments, View Attendance, View Marks & Grades, View Current CGPA, Placement Drives, Notice Board, Give Feedback.

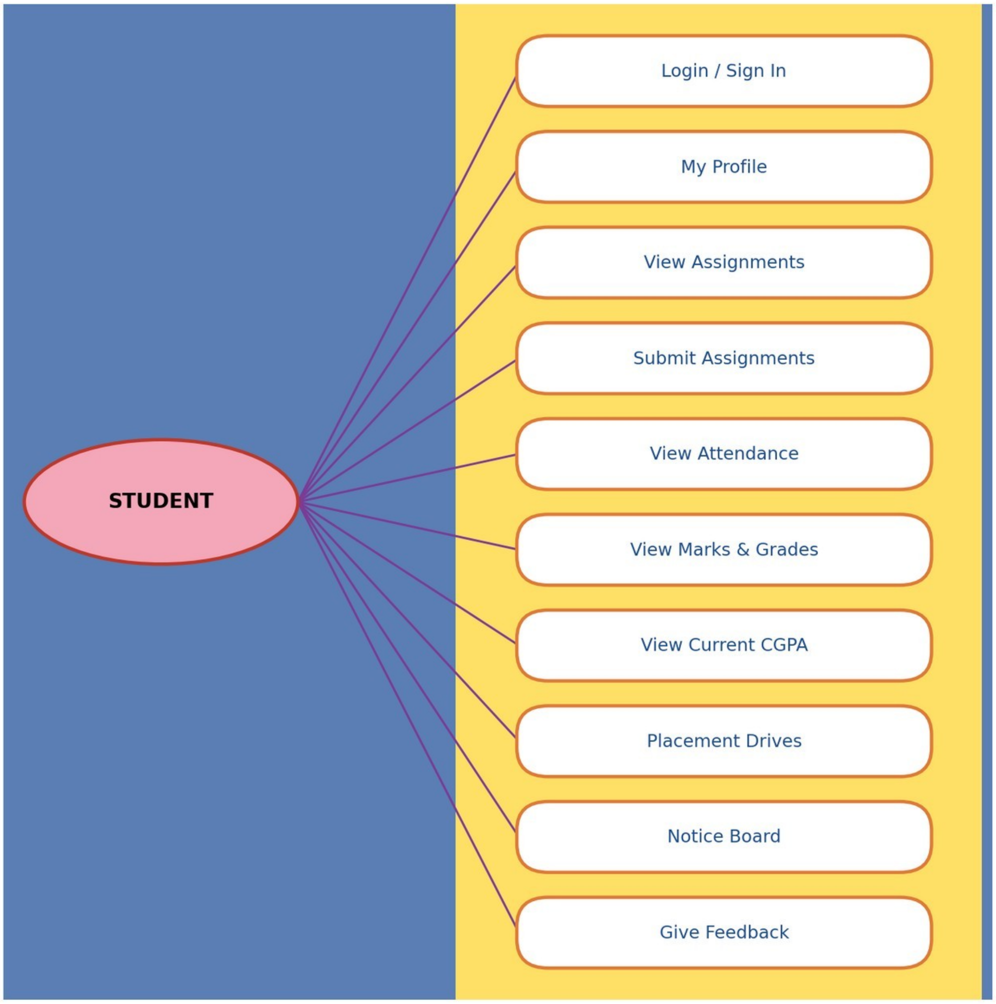

Fig 4.1: Use-case diagram — Student

4.1.2 Complete System — Staff / Faculty Module

Staff/Faculty use cases: Login/Sign In, Manage Students, Manage Marks & Grades, Manage Assignments, Manage Attendance, Manage Course Modules, Manage Curriculum, Manage Placement Drives, Post Notifications, View Student Feedback.

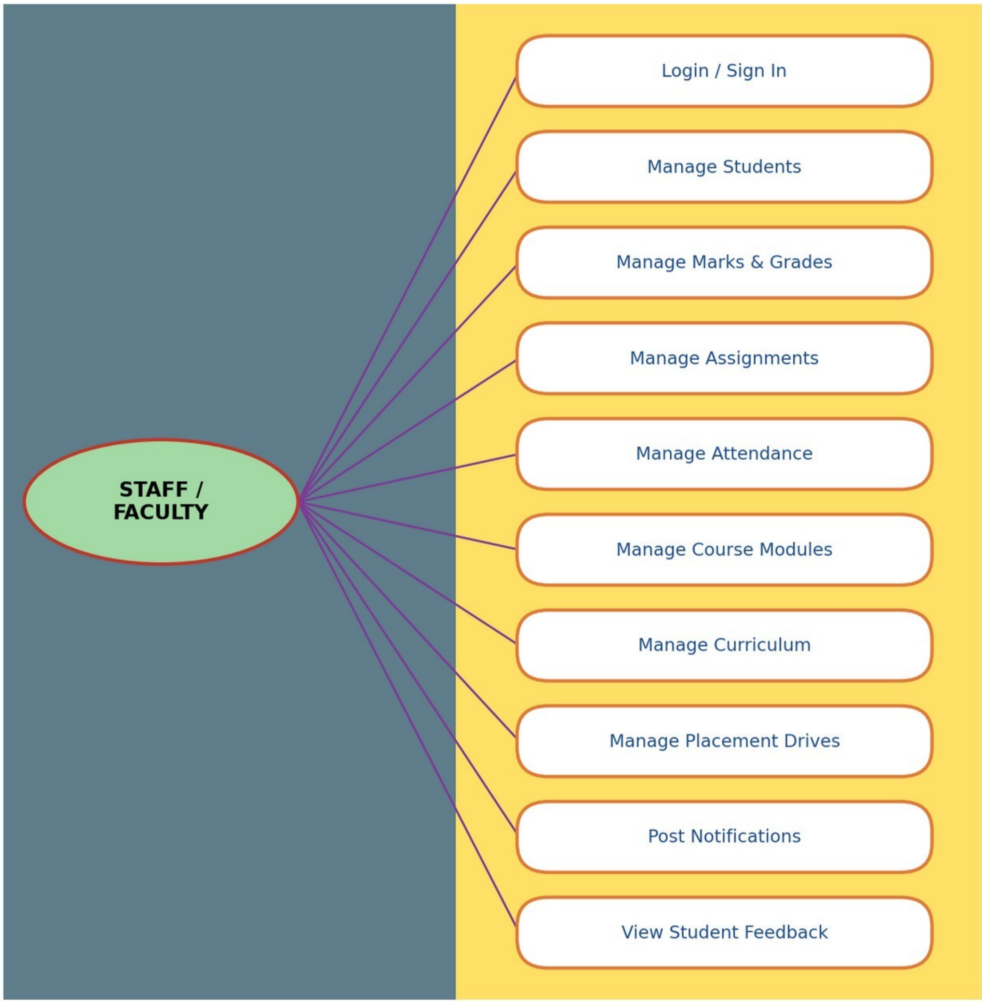

Fig 4.2: Use-case diagram — Staff/Faculty

4.2 User Interface Requirements

The system provides the following primary navigation for each role:

Student navigation: Home | My Profile | Assignments | Attendance | Marks | Placements | Feedback | Notice Board

Staff navigation: Home | Students | Grades | Assignments | Attendance | Feedback | Placement | Notifications

5. Design

5.1 High Level Design

The CDAC Student Management System follows a 3-tier architecture:

- Presentation Layer — React + Bootstrap single-page application, providing separate Student and Staff dashboards.
- Application/Service Layer — Spring Boot REST services for core academic modules (students, marks, grades, attendance, assignments, placements) and Node.js REST services for notice board, notifications and feedback.
- Data Layer — MySQL relational database storing all persistent records.

The React front end authenticates users via their institute email and password, then makes secure REST calls to the appropriate back-end service based on the module being accessed (Spring Boot for academic data, Node.js for communication features).

5.2 Database Design

The following ER diagram depicts the database design used by the system, covering student, staff, course, module, attendance, marks, assignment, assignment submission, placement drive, notice, feedback, and centre entities.

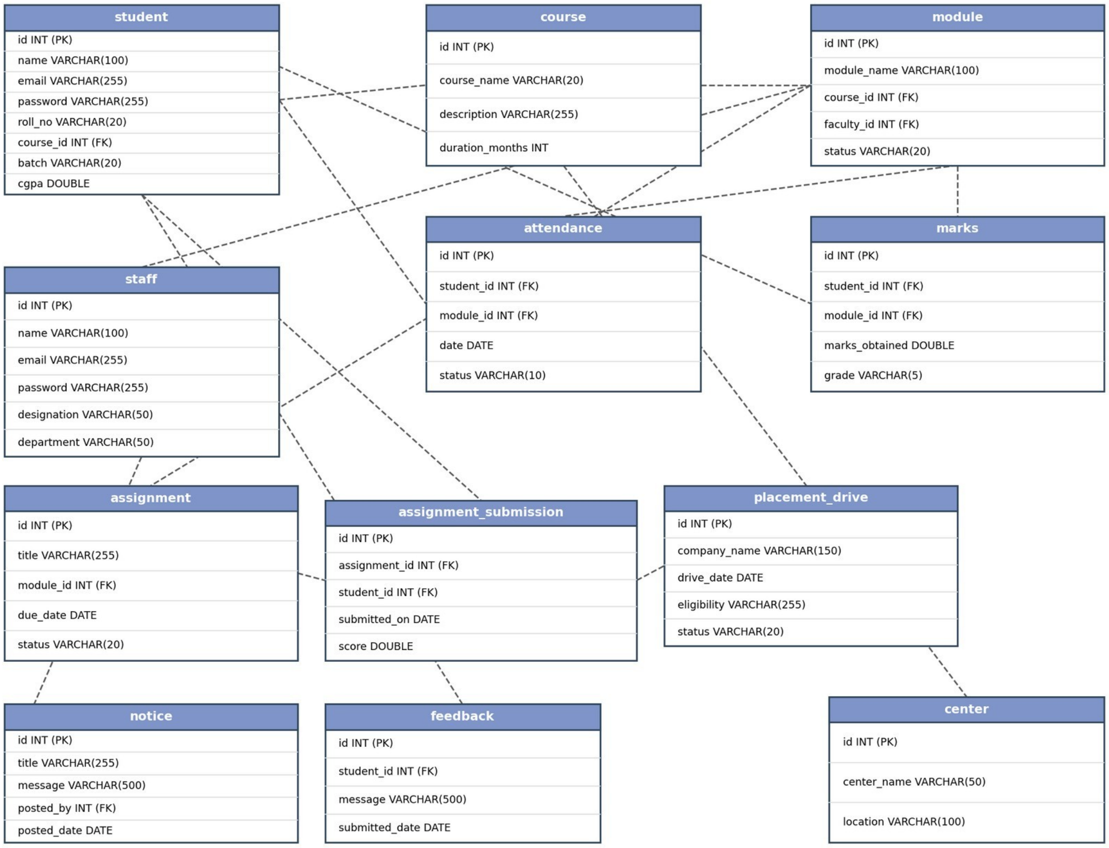

Fig 5.1: Entity-Relationship (ER) Diagram

6. Interface (UI)

Login Page

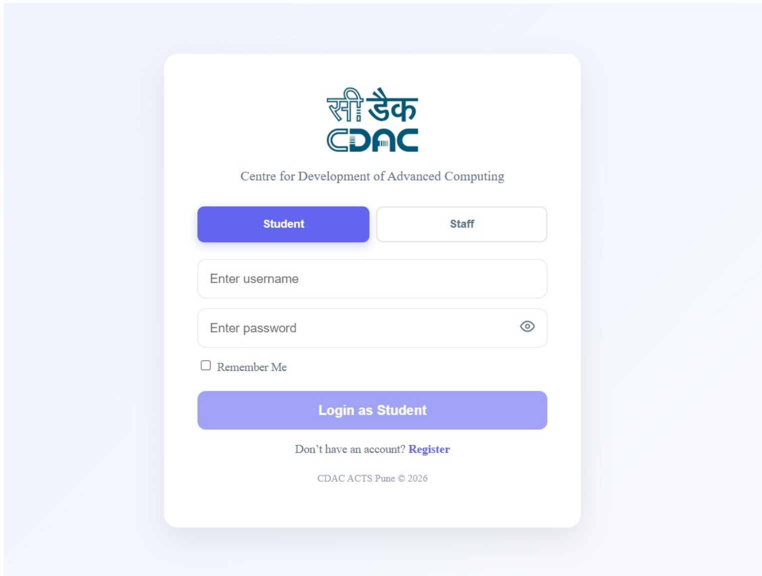

Fig 6.1: Login Page — role-based sign in for Student / Staff

Student Dashboard

On successful login, students land on a dashboard summarizing current CGPA, attendance standing, modules cleared, and active assignments, along with the notice board, upcoming assignments, and open placement drives.

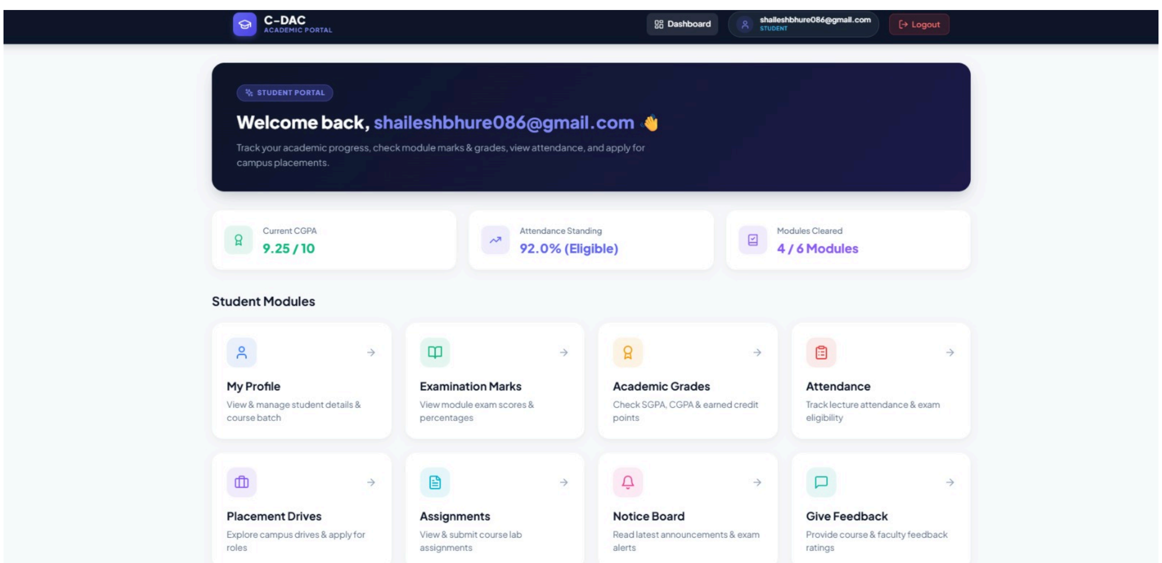

Fig 6.2: Student Dashboard

Staff / Faculty Dashboard

Staff members land on a dashboard summarizing enrolled students, active placement drives, average attendance, and courses offered, along with course-wise enrollment, curriculum details, and a student management table.

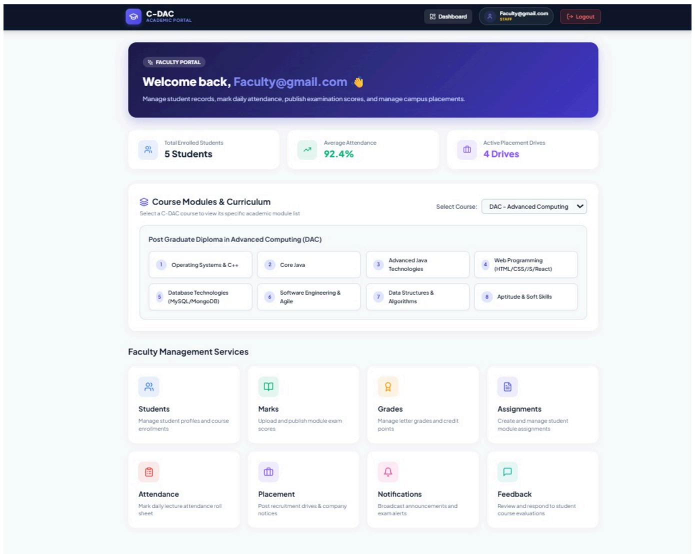

Fig 6.3: Staff Dashboard

7. Test Report

The report of the testing carried out on the CDAC Student Management System is given below.

Sr. No	Test Case Title	Description	Expected Outcome	Error Message	Result
1	Login Page	User should see the login page and select role (Student/Staff), then enter email and password.	After signing in, user is directed to the respective dashboard.	Invalid Login	Passed
2	Student Dashboard Displayed	Dashboard is displayed for every successful student login.	CGPA, attendance, modules cleared and assignments are displayed.	No Error	Passed
3	My Profile	Student can view and update personal/academic profile details.	Profile details displayed correctly.	No Error	Passed
4	Assignments	Student can view assignments and submit work before due date.	Assignment submitted and status updated.	Submission Error	Passed
5	Attendance	Student can view module-wise and overall attendance percentage.	Attendance records displayed accurately.	No Error	Passed
6	Marks & Grades	Student can view marks and	Marks/grades displayed	No Error	Passed

Sr. No	Test Case Title	Description	Expected Outcome	Error Message	Result
		grades for each module.	correctly.		
7	Placement Drives	Student can view active and closed placement drives with eligibility.	Drive list displayed with correct status.	No Error	Passed
8	Notice Board	Student/Staff can view posted notices.	Notices displayed in chronological order.	No Error	Passed
9	Feedback	Student can submit feedback; staff can view submitted feedback.	Feedback recorded and visible to staff.	Validation Error	Passed
10	Manage Students (Staff)	Staff can view/manage the list of enrolled students.	Student list displayed with attendance and CGPA.	No Error	Passed
11	Manage Marks/Grades (Staff)	Staff can enter/update marks and grades for a module.	Marks and grades saved successfully.	Validation Error	Passed
12	Manage Assignments (Staff)	Staff can create assignments and evaluate submissions.	Assignment created/evaluated successfully.	No Error	Passed
13	Manage Attendance (Staff)	Staff can mark daily/module-wise attendance for students.	Attendance saved successfully.	No Error	Passed
14	Manage Course Modules & Curriculum (Staff)	Staff can add/update course modules and curriculum content.	Modules updated successfully.	No Error	Passed
15	Manage Placement Drives (Staff)	Staff can create and update placement drive details.	Placement drive saved and visible to eligible students.	No Error	Passed
16	Sign Out	User should be able to log out from the portal.	User logged out and redirected to the login page.	No Error	Passed

8. Project Management Methodology

The Scrum Agile methodology was used for the development of the CDAC Student Management System, with work divided into sprints covering the student module, staff module, database design, and integration/testing phases.

9. Future Scope

- Integrate an online fee payment gateway for course fees.
- Add an SMS/email-based automated notification system for attendance shortage and placement drive alerts.
- Introduce a mobile application (Android/iOS) version of the portal for students and staff.
- Add analytics and reporting dashboards for centre-wide performance across DAC, DBDA, DESD and DITISS batches.

- Integrate video-based online class attendance and recorded lecture access.
- Add resume-building and mock-interview scheduling features for placement preparation.

10. Conclusion

The CDAC Student Management System successfully brings together student records, attendance, marks, assignments, and placement activity into a single, secure web platform for C-DAC Hyderabad (Ameerpet). By separating the Student and Staff/Faculty experiences behind role-based login, the system gives students self-service visibility into their own academic progress while giving faculty a consolidated dashboard to manage day-to-day academic administration across the DAC, DBDA, DESD and DITISS courses.

The use of React and Bootstrap on the front end, combined with Spring Boot and Node.js REST services and a MySQL relational database on the back end, ensured a responsive, modular, and maintainable architecture. All sixteen test cases covering login, dashboards, assignments, attendance, marks, placements, notifications, and staff administration passed successfully, confirming that the core functional requirements outlined in this report have been met.

Overall, the project met its stated objectives of automating manual academic record-keeping and improving communication between students and staff. The future scope items identified — including payment gateway integration, automated notifications, a mobile application, and analytics dashboards — provide a clear roadmap for extending the system beyond this initial release.